

Vascular proliferations and abnormalities of blood vessels

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Introduction

The terminology of vascular conditions can be confusing, with several lesions being incorrectly named or classified. Vascular skin lesions include:

Vascular naevi

Angiomas

Telangiectases

Malignant vascular tumours

A paediatric dermatologist, paediatrician, vascular specialist or surgeon should assess significant infantile [capillary malformations](#) or infantile haemangiomas, especially when large, symptomatic (especially when ulcerated), located on the head and neck or close to eye, nose or mouth.

Vascular naevus

Vascular naevi or anomalies are present at birth or appear in early childhood. They are classified according to the size and type of vessel. They may remain stable or become more prominent with maturity. There are various associated syndromes.

[Capillary malformation](#) – [salmon patch](#) and port wine stain.

[Macrocephaly-capillary malformation](#) and other [PIK3CA-related overgrowth spectrum](#) disorders

[Venous malformation](#) – [glomovenous malformation](#), arteriovenous malformations, blue rubber bleb syndrome (associated gastrointestinal lesions).

Arteriovenous malformations: of mixed blood vessel origin

[Lymphatic malformation](#): lymphangioma circumscriptum, cavernous lymphangioma

[Angioma serpiginosa](#): a crop of swirling vascular papules

[Naevus anaemicus](#) – in this naevus, blood flow is interrupted resulting in pale areas.



Capillary vascular malformation



Lymphatic malformation



Naevus anaemicus

Angioma

An angioma or haemangioma is a benign tumour formed by the dilation of blood vessels or the formation of new ones by the proliferation of endothelial cells.

Infantile haemangioma (superficial, deep, or mixed): proliferates in the first few weeks of life, followed by involution later in childhood. These are sometimes syndromic, as in **PHACE**, **LUMBAR** and **PELVIS** syndromes.

Glomeruloid haemangioma is an eruptive form of haemangioma typically diagnosed in **POEMS** syndrome

Congenital haemangioma is at full size at birth and may rapidly involute (**RICH**) or persist (**NICH**)

Kaposiform haemangioendothelioma is a rare aggressive haemangioma that results in platelet trapping — the **Kasabach-Merritt phenomenon**.

Tufted angioma: rare childhood tumour with characteristic histology.

Cherry angioma: adult onset, common degenerative lesions, usually multiple

Angiokeratoma: acquired scaly angiomas, usually on vulva or scrotum, or in association with **Fabry disease**

Glomus tumour: tender papule on nail bed or palm arising in young to middle-aged adult

Pyogenic granuloma occurs in children and young adults on skin and mucosa, most often lower lips, fingers and toes. It may develop in response to a minor injury, in pregnancy, or when taking certain medications. It grows rapidly and may become pedunculated or polypoid and surrounded by a collarette of normal skin. It bleeds and crusts.

Bacillary angiomatosis is a rare opportunistic bacterial infection due to *Bartonella henselae*.



Infantile haemangioma



Cherry angiomas



Venous lake

Telangiectasia

Prominent cutaneous blood vessels can be physiological or pathological.

Blood vessels feeding a tumour such as [basal cell carcinoma](#)

A common sign of [rosacea](#)

A diagnostic feature of the CRST variant of [systemic sclerosis](#)

There are some named conditions in which telangiectasia is (or telangiectases are) characteristic.

Telangiectasia eruptiva macularis perstans, a form of [mastocytosis](#)

[Spider telangiectasis](#) consists of central arteriole and radiating capillaries. Very common in healthy individuals, but more arise in response to oestrogen, often in pregnancy, liver disease.

[Venous lake](#): blue or purple compressible papule due to venous dilation, often on lower lip or ear.

Unilateral acquired telangiectasia: telangiectasia with naevoid distribution.

[Essential telangiectasia](#): idiopathic telangiectasia and venulectasia.

[Benign hereditary telangiectasia](#): a familial condition in which many telangiectases appear

[Hereditary haemorrhagic telangiectasia](#): telangiectasia on the skin and mucous membranes associated with bleeding from nose and gut causing anaemia.

[Diffuse dermal angiomatosis of the breast](#): painful ulceration and telangiectases of pendulous breasts

Acquired [lymphangiectasia](#): "frog-spawn" appearance that follows lymphatic obstruction eg tumour or surgery.



Spider telangiectasis



Essential telangiectasis



Hereditary haemorrhagic telangiectasis

Malignant vascular tumours

Kaposi sarcoma: four types are recognised due to Human herpesvirus 8 (HHV8). Multiple purple macules, papules and plaque on the skin and mucous membranes develop.

Angiosarcoma: idiopathic or secondary to chronic lymphoedema or radiation. Often aggressive, it mostly presents in elderly people with spreading purple patches and plaques that may bleed and ulcerate.

Intravascular B-cell lymphoma presents with rapid-onset irregular telangiectases.



Kaposi sarcoma



Angiosarcoma



Intravascular lymphoma